

Hoang Trieu

+1 813-204-0057 | hoangtrieuhuynh22@augustana.edu | [LinkedIn](#) | [Website](#)

EDUCATION

Augustana College

Mathematics and Computer Science Double Major

Rock Island, Illinois

Feb. 2023 – Present

- GPA: **3.8/4.0**, Dean's List, Pi Mu Epsilon (National Mathematics Honor Society)
- **Relevant coursework:** Functional Analysis, Partial Differential Equations, Abstract Algebra, Algorithm and Computational Theory, Data Structures, Software Development

Budapest Semester in Mathematics

Semester Exchange

Budapest, Hungary

Jan. 2025 - May 2025

- **Relevant coursework:** Real Analysis, Topology (Algebraic, Geometric, and Point Set), Deep Learning, Combinatorics I (Enumerative Combinatorics and Graph Theory)

DIRECTED STUDY PROGRAM

Topics in Functional Analysis

Reading Course with Dr. Benjamin Civiletti (Augustana College)

Rock Island, Illinois

Sept. 2025 – Present

- Study foundational principles of Functional Analysis, covering Banach and Hilbert space theory, key theorems (Hahn-Banach, Open Mapping, Riesz Representation), Measure Theory, and applications to Fourier Series.

Topics in Topology

Reading Program with Szemerédi Levente (Budapest Semester in Math)

Budapest, Hungary

Jan. 2025 - May 2025

- Studied topological properties of function spaces by applying the Baire Category Theorem to prove the density of the set of nowhere-differentiable continuous functions.
- Resulted in a student's write-up, which can be found [here](#).

Topics in Non-linear Differential Equation

Beling Scholar Program with Dr. Andrew Sward (Augustana College)

Rock Island, Illinois

Sept. 2023 - Dec. 2023

- Studied the properties of nonlinear differential equations in Physics models, particularly in the Predator-Prey model where the predator has a constant acceleration. Discovered an approximation of the solution using power series and recurrence relation.

MANUSCRIPT IN PREPARATION

Closed Combinatorial and Trigonometry Expansion of $(1 - \zeta_p)^n$, with Randazzo (Augustana College)

Developed two explicit formulas for the coefficients in the expansion of $(1 - \zeta_p)^n$ using roots of unity, providing new analytical tools that offer potential progress toward solving an open problem on conductors in specific rings

RESEARCH EXPERIENCE

Undergraduate Student Researcher - Dr. Brooke Randazzo

Research Topic: Closed Combinatorial and Trigonometry Expansion of $(1 - \zeta_p)^n$

June. 2025 - Present

Rock Island, IL

- **Derived two novel closed-form formulas** for the coefficients of the expansion $(1 - \zeta_p)^n$ of using both the Binomial Theorem and the Root of Unity Filter technique, contributing potential tools toward solving an open problem in algebraic number theory.
- **Implemented simulations and code** to model and visualize the behavior of the coefficient formulas, validating theoretical results.
- **Authored** the manuscript and conducted literature review.

Research Assistant - Dr. Benjamin Civiletti

Feb. 2024 - Dec. 2024

Research Project: Numerical Method for Electro-optics Optimization

Rock Island, IL

- **Implemented in Python the hybrid C-RCWA algorithm** that mitigated Gibbs phenomenon for Maxwell solver near discontinuities
- **Optimized code maintainability** by refactoring code into an object-oriented architecture, following common design patterns for modularity and clarity
- **Validated model and improved accuracy** by writing test cases and ran simulations

SEMINAR TALK AND PRESENTATION

(2024) **Coordinate Transform Method for Solving Maxwell's Equations** (Augustana College)

(2024) **An Interactive-based Workshop on Maxwell's equations Solver** (Augustana College)

(2023) **Pursuit Curve for An Accelerating Target – A Closer Look** (Augustana College)

TEACHING AND TUTOR EXPERIENCE

Individual Tutor and Supplemental Instruction Leader

Sept. 2024 – Present

Augustana College - Learning Commons

Rock Island, IL

- Mentor undergraduate students in 7+ core mathematics courses, including Pre-calculus, Calculus I, II, III, Differential Equations, Discrete Math, Real Analysis, Abstract Algebra.
- Develop and lead weekly Supplemental Instruction (SI) sessions for College Algebra and Multivariable Calculus to reinforce complex concepts.
- Elevated student performance in College Algebra (Fall 2024), with SI participants achieving **18.8%** higher homework scores and a **10.1 percentage point** higher midterm exam average.
- Boosted student mastery in Multivariable Calculus (Fall 2025), driving a **14%** average increase in quiz scores for session participants, with individual improvements as high as **69.2%**.

SIMULATION PROJECT

HAM Radio Simulator | *Java, Python, JavaFX, Google Cloud, Gemini API*

Sept. 2024 - Dec. 2024

- **Developed a multi-user networking radio simulator** using Google Cloud Platform, enabling real-time continuous wave (CW) signal exchange between clients with encoded metadata.
- **Designed and integrated AI-driven chatbots** into the HAM Radio simulator using Gemini API, creating persona-based agents with customizable prompts to enhance user interaction and engagement.

RESEARCH-RELATED SKILLS

Programing Language: Python (with SciPy, Numpy, Matplotlib, PyTorch FENICs), Java, Matlab, R

Numerical Method: Finite Element Method, Finite-Difference Time-Domain, Finite-Difference

Frequency-Domain, Rigorous-Coupled Wave Approach (RCWA), Fast-Fourier Transform